



Symbiosis Institute of Technology, Pune

CSE- Academic Year 2026-27

Java Programming – Lab Batch 2025-29

Lab Assignment No.: 14	
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Batch	2025-2029
Class	A1
Academic Year & Semester	Second Year Semester III
Date of Submission	
Title of Assignment:	Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks.
Problem Statement & Learning Objective	Problem Statement Write a Java program to demonstrate exception handling using try and catch blocks, nested and multiple try-catch blocks. Learning Objectives To handle runtime exceptions using try-catch, nested try, and multiple catch blocks.
Theory	Exception handling prevents abnormal termination of programs due to runtime errors. The try block contains code that may generate exceptions, while catch blocks handle specific exceptions. Nested and multiple try-catch structures allow complex error management. Exception handling improves software reliability and user experience by gracefully managing unexpected situations.

Program Code**Main1: (0 case)**

```
public class Main1 {  
    public static void main (String[] args) {  
        int num1 = 10;  
        int num2 = 0;  
  
        try {  
            int result = num1 / num2;  
            System.out.println("Result : " + result);  
        } catch (ArithmeticException e) {  
            System.out.println("Error : Cannot divide by  
zero.");  
            System.out.println("Error : " + e.getMessage());  
        }  
    }  
}
```

```
        System.out.println("Program continues after exception  
handling.");  
    }  
}
```

Main2: (Error Display)

```
public class Main2 {  
    public static void main (String[] args) {  
        try {  
            String str = "123a";  
            int num = Integer.parseInt(str);  
  
            int r = 10 / 0;  
            System.out.println(num + r);  
        } catch (NumberFormatException | ArithmeticException e) {  
            System.out.println("Error : " +  
e.getClass().getSimpleName() + " occurred.");  
        }  
    }  
}
```

Main3: (Divide by 0 case)

```
public class Main3 {  
    public static void main (String[] args) {  
        try {  
            System.out.println("Outer try block started");  
  
            try {  
                System.out.println("Inner try block started");  
                int result = 10 / 0;  
                System.out.println(result);  
            } catch (ArithmeticException e) {  
                System.out.println("Error + " + e.getMessage());  
            }  
        }  
    }  
}
```

	<pre> } int result = 10 / 0; System.out.println(result); } catch (NullPointerException e) { System.out.println("Caught NullPointerException in outer try block : " + e.getMessage()); } finally { System.out.println("Finally block of outer try executed"); } } System.out.println("Program continues after nested try- catch blocks"); } } </pre>
Output Screenshots	Main1: <pre> ● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 14> & m\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCodeDetail sers\Nilima\AppData\Roaming\Code\User\workspaceStorage\204 hat.java\jdt_ws\LAB EXP 14_cc80a22b\bin' 'Main1' Error : Cannot divide by zero. Error : / by zero Program continues after exception handling. ○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 14> </pre> Main2: <pre> ● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 14> & ' m\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCodeDetails sers\Nilima\AppData\Roaming\Code\User\workspaceStorage\2048 hat.java\jdt_ws\LAB EXP 14_cc80a22b\bin' 'Main2' Error : NumberFormatException occurred. ○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 14> </pre> Main3: <pre> ● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 14> & 'c:\Program m\jdk-24.0.2.12-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInException sers\Nilima\AppData\Roaming\Code\User\workspaceStorage\2048ff43486d8b9 hat.java\jdt_ws\LAB EXP 14_cc80a22b\bin' 'Main3' Outer try block started Inner try block started Error + / by zero Finally block of outer try executed Exception in thread "main" java.lang.ArithmeticException: / by zero at Main3.main(Main3.java:15) ❖ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 14> </pre>

Exercise Qns and Code

1. Create an ATM program handling invalid withdrawal amount using try-catch

```
import java.util.Scanner;

public class Ex1 {
    public static void main(String[] args) {
        try (Scanner in = new Scanner(System.in)) {
            double balance = 10000;

            try {
                System.out.print("Enter withdrawal amount: ");
                double amount = in.nextDouble();

                if (amount <= 0) {
                    throw new Exception("Invalid withdrawal amount!");
                }

                if (amount > balance) {
                    throw new Exception("Insufficient balance!");
                }

                balance -= amount;

                System.out.println("Withdrawal Successful.");
                System.out.println("Remaining Balance: Rs." + balance);

            } catch (Exception e) {
                System.out.println("Error: " + e.getMessage());
            }
        }
    }
}
```

2. Create an Online Shopping program that handles an invalid product quantity entered by the user using a try-catch block. Display an appropriate error message if the quantity is less than or equal to zero.

```
public class Exp2 {
    public static void main (String[] args) {
        int products = 10;

        try (Scanner in = new Scanner(System.in)) {
            try {
                System.out.print("Enter number of products to but : ");

                int am = in.nextInt();

                if (am < 0) {
                    throw new Exception("Invalid Product Amount.");
                }
            }
        }
    }
}
```

}

```
        if (am == 0) {
            throw new Exception("Please enter amount
more than 0.");
        }
```

```
        if (am > products) {
            throw new Exception("Not enough products in
stock to buy.");
        }
    }
}
```

```

        System.out.println("Purchased!");
        products -= am;
        System.out.println("Remaining Products : " +
products);
    } catch (Exception e) {
        System.out.println("Error: " + e.getMessage());
    }
}
}
}
}

```

Output Screenshots

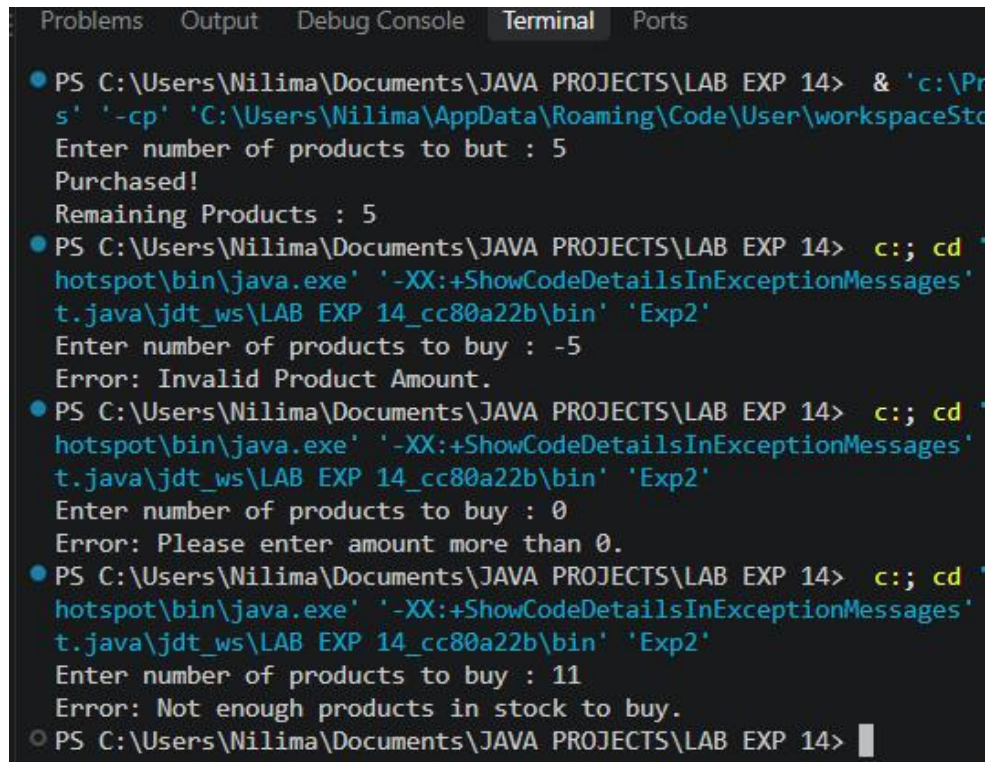
Ex1:

```
PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 14> & 'c:\
-hotspot\bin\java.exe' '-XX:+ShowCodeDetailsInExceptionMessage
de\User\workspaceStorage\2048ff43486d8b98e67880dfe3f9dbc1\rec
'

Enter withdrawal amount: 4000
Withdrawal Successful.
Remaining Balance: Rs.6000.0

PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 14>
```

Ex2: (all cases)



```
Problems  Output  Debug Console  Terminal  Ports

● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 14> & 'c:\Pr
s' '-cp' 'C:\Users\Nilima\AppData\Roaming\Code\User\workspaceSto
Enter number of products to but : 5
Purchased!
Remaining Products : 5
● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 14> c:; cd "
hotspot\bin\java.exe" '-XX:+ShowCodeDetailsInExceptionMessages'
t.java\jdt_ws\LAB EXP 14_cc80a22b\bin' 'Exp2'
Enter number of products to buy : -5
Error: Invalid Product Amount.
● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 14> c:; cd "
hotspot\bin\java.exe" '-XX:+ShowCodeDetailsInExceptionMessages'
t.java\jdt_ws\LAB EXP 14_cc80a22b\bin' 'Exp2'
Enter number of products to buy : 0
Error: Please enter amount more than 0.
● PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 14> c:; cd "
hotspot\bin\java.exe" '-XX:+ShowCodeDetailsInExceptionMessages'
t.java\jdt_ws\LAB EXP 14_cc80a22b\bin' 'Exp2'
Enter number of products to buy : 11
Error: Not enough products in stock to buy.
○ PS C:\Users\Nilima\Documents\JAVA PROJECTS\LAB EXP 14> |
```

Conclusion

This experiment covers runtime error management to ensure program stability. By utilizing try-catch blocks and ordered multiple catch structures, it illustrates how to intercept specific runtime anomalies and prevent sudden program termination during invalid operational inputs.